Disjoint Set Union

September 6, 2026

A (ID 2200110)

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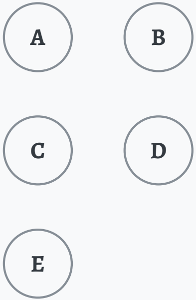
C (ID 2200110)

Department of Computer Science and Engineering

Outline

Introduction

A Familiar Problem



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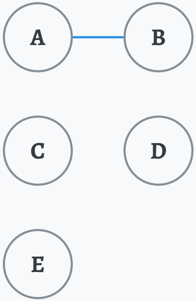

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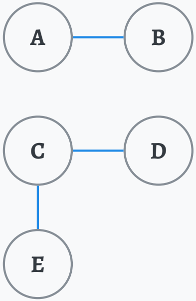

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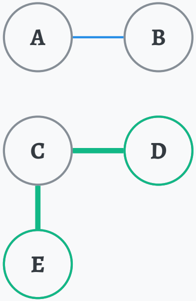

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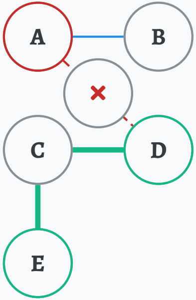

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- What about A and D? No — they belong to two different circles entirely.

Larger Networks

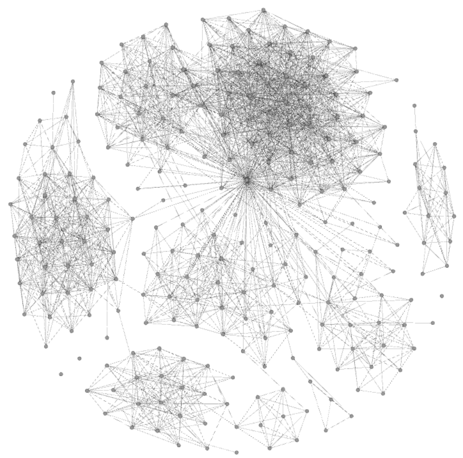

Figure: *Social Media Network Visualization*[a]

- Consider a large social network, or equally, a large computer network, with countless members.

[a] Pinterest: Visualise Your Facebook Friend Network

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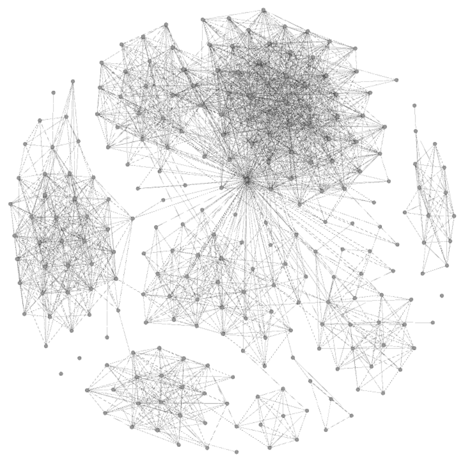

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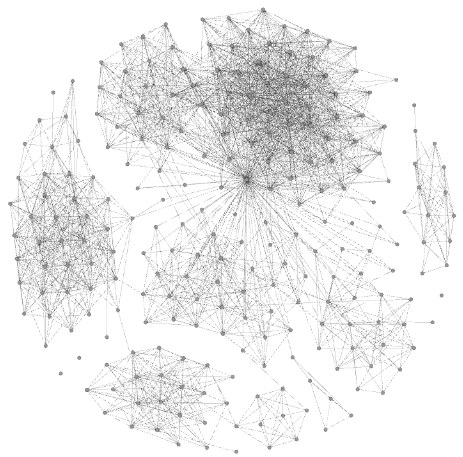

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This is exactly what *Disjoint Set Union* answers.

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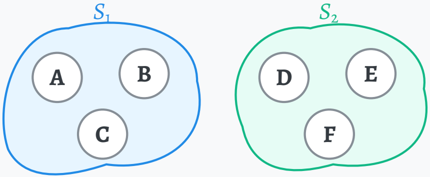
What Is a Disjoint Set Union?



Definition

- Sets are disjoint if they share no elements:

$$S_i \cap S_j = \varnothing \quad \text{for } i \neq j.$$

- **Disjoint Set Union** or **DSU** are a collection of such disjoint groups.

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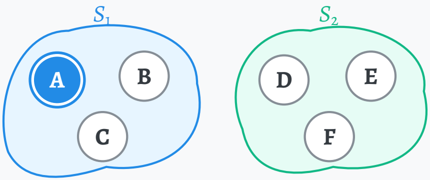

Here,

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- One chosen element identifies the entire set.

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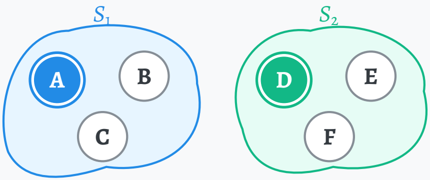

Here,

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- D is the representative of S_2

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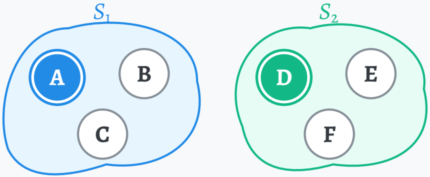

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Same representative $\Longleftrightarrow$ same set.